Right-Triangle Trigonometry

Trigonometric ratios

1 A right triangle has legs of length 5 and 12 and hypotenuse 13. For the angle θ opposite the side of length 5, find:

a $\sin\theta$ **b** $\cos\theta$ **c** $\tan\theta$

2 Write the exact value of each expression:

a $\sin 30°$ **b** $\cos 45°$ **c** $\tan 60°$ **d** $\sin 60°$

3 A right triangle has legs of length 7 and 24.

a Find the length of the hypotenuse.

b Find the measure of the angle opposite the side of length 7, to the nearest hundredth of a degree.

Applications

4 A ladder leans against a wall, making an angle of $62°$ with the ground. The foot of the ladder is 4.5 m from the base of the wall. How high up the wall does the ladder reach? Round to two decimal places.

5 From the top of a cliff 50 m above sea level, the angle of depression to a boat is $18°$. How far is the boat from the base of the cliff? Round to one decimal place.

6 A kite string is 85 m long and makes an angle of $54°$ with the horizontal ground. Assuming the string is straight, find the height of the kite above the ground, to the nearest metre.